

Supplementary Material

Diffusion-based Frameworks for Unsupervised Speech Enhancement

This supplementary material provides additional experimental results and analyses supporting the main manuscript. It includes average metrics with uncertainty measures, results by noise type and SNR, performance–speed trade-off analyses, complementary violin plots, and preliminary inference-time adaptation experiments.

Contents

A	Additional results on the test datasets	2
A.1	Average metrics with uncertainty measures	2
A.2	Average metrics by noise type	4
A.3	Average metrics by noise type and SNR	9
B	Effect of speech and noise mismatch	13
C	Performance–speed trade-off of DiffUSEEN and DEPSE-TL	15
D	Metric distributions in matched and mismatched scenarios	16
E	Preliminary trials on inference-time adaptation for the diffusion noise model in ParaDiffUSE-EN	19

A Additional results on the test datasets

This section reports additional quantitative results on the WSJ0-QUT and VB-DMD test sets. We first provide average metrics with uncertainty measures, then report results by noise type, and finally decompose the results by both noise type and SNR.

A.1 Average metrics with uncertainty measures

We decompose the main results table of the paper into smaller tables to improve readability and include the standard error of each reported average.

Table 1: Average performance on WSJ0-QUT under matched conditions. Values are reported as mean $\pm$ standard error.

	SI-SDR $\pm$ SE	SI-SIR $\pm$ SE	SI-SAR $\pm$ SE	PESQ $\pm$ SE	ESTOI $\pm$ SE	DNS-MOS $\pm$ SE
Input	-2.60 $\pm$ 0.16	-2.60 $\pm$ 0.16	46.66 $\pm$ 0.43	1.83 $\pm$ 0.02	0.50 $\pm$ 0.01	2.69 $\pm$ 0.01
SGMSE+	8.53 $\pm$ 0.17	24.65 $\pm$ 0.29	8.72 $\pm$ 0.16	2.61 $\pm$ 0.02	0.76 $\pm$ 0.00	3.63 $\pm$ 0.01
Conv-TasNet	12.63 $\pm$ 0.14	29.88 $\pm$ 0.26	12.75 $\pm$ 0.14	2.86 $\pm$ 0.02	0.83 $\pm$ 0.00	3.57 $\pm$ 0.01
RemixIT	4.14 $\pm$ 0.07	22.45 $\pm$ 0.33	4.32 $\pm$ 0.06	2.21 $\pm$ 0.01	0.68 $\pm$ 0.00	3.06 $\pm$ 0.01
RVAE	6.22 $\pm$ 0.21	14.18 $\pm$ 0.33	7.7 $\pm$ 0.18	2.33 $\pm$ 0.02	0.64 $\pm$ 0.01	3.25 $\pm$ 0.01
UDiffSE	4.35 $\pm$ 0.22	9.63 $\pm$ 0.29	6.24 $\pm$ 0.19	2.20 $\pm$ 0.02	0.61 $\pm$ 0.01	3.06 $\pm$ 0.01
UDiffSE+	3.24 $\pm$ 0.21	12.14 $\pm$ 0.35	4.23 $\pm$ 0.18	2.21 $\pm$ 0.02	0.58 $\pm$ 0.01	3.19 $\pm$ 0.01
DEPSE-IL	3.25 $\pm$ 0.21	9.63 $\pm$ 0.32	4.85 $\pm$ 0.17	2.20 $\pm$ 0.02	0.59 $\pm$ 0.01	3.11 $\pm$ 0.02
DEPSE-TL	3.53 $\pm$ 0.22	5.72 $\pm$ 0.26	8.05 $\pm$ 0.17	2.17 $\pm$ 0.02	0.61 $\pm$ 0.01	3.04 $\pm$ 0.02
DiffUSEEN	5.37 $\pm$ 0.22	9.69 $\pm$ 0.30	7.97 $\pm$ 0.19	2.33 $\pm$ 0.02	0.67 $\pm$ 0.01	3.16 $\pm$ 0.01
ParaDiffUSE-IN	2.02 $\pm$ 0.24	4.86 $\pm$ 0.31	6.42 $\pm$ 0.17	2.16 $\pm$ 0.02	0.56 $\pm$ 0.01	3.24 $\pm$ 0.01
ParaDiffUSE-EN	8.49 $\pm$ 0.19	21.54 $\pm$ 0.30	8.90 $\pm$ 0.18	2.61 $\pm$ 0.02	0.73 $\pm$ 0.01	3.44 $\pm$ 0.02

Table 2: Average performance on WSJ0-QUT under mismatched conditions. Values are reported as mean $\pm$ standard error.

	SI-SDR $\pm$ SE	SI-SIR $\pm$ SE	SI-SAR $\pm$ SE	PESQ $\pm$ SE	ESTOI $\pm$ SE	DNS-MOS $\pm$ SE
Input	-2.6 $\pm$ 0.16	-2.6 $\pm$ 0.16	46.66 $\pm$ 0.43	1.83 $\pm$ 0.02	0.5 $\pm$ 0.01	2.69 $\pm$ 0.01
SGMSE+	3.02 $\pm$ 0.28	12.79 $\pm$ 0.4	3.86 $\pm$ 0.26	2.08 $\pm$ 0.03	0.61 $\pm$ 0.01	3.59 $\pm$ 0.01
Conv-TasNet	5.83 $\pm$ 0.2	17.07 $\pm$ 0.31	6.36 $\pm$ 0.19	2.21 $\pm$ 0.02	0.63 $\pm$ 0.01	3.07 $\pm$ 0.01
RemixIT	3.35 $\pm$ 0.07	19.88 $\pm$ 0.34	3.62 $\pm$ 0.06	2.13 $\pm$ 0.01	0.65 $\pm$ 0.0	2.94 $\pm$ 0.01
RVAE	4.98 $\pm$ 0.2	11.66 $\pm$ 0.29	6.75 $\pm$ 0.18	2.17 $\pm$ 0.02	0.59 $\pm$ 0.01	3.1 $\pm$ 0.01
UDiffSE	1.66 $\pm$ 0.24	8.51 $\pm$ 0.36	3.14 $\pm$ 0.21	2.1 $\pm$ 0.02	0.56 $\pm$ 0.01	3.07 $\pm$ 0.01
UDiffSE+	0.06 $\pm$ 0.25	8.72 $\pm$ 0.41	1.21 $\pm$ 0.21	2.05 $\pm$ 0.02	0.52 $\pm$ 0.01	3.13 $\pm$ 0.01
DEPSE-IL	0.24 $\pm$ 0.25	6.63 $\pm$ 0.37	1.91 $\pm$ 0.2	2.06 $\pm$ 0.02	0.53 $\pm$ 0.01	3.05 $\pm$ 0.02
DEPSE-TL	2.66 $\pm$ 0.24	5.96 $\pm$ 0.3	6.07 $\pm$ 0.19	2.16 $\pm$ 0.02	0.59 $\pm$ 0.01	3.05 $\pm$ 0.01
DiffUSEEN	4.59 $\pm$ 0.22	11.79 $\pm$ 0.36	6.15 $\pm$ 0.19	2.31 $\pm$ 0.02	0.64 $\pm$ 0.01	3.23 $\pm$ 0.01
ParaDiffUSE-IN	-0.39 $\pm$ 0.27	3.18 $\pm$ 0.33	2.78 $\pm$ 0.22	1.98 $\pm$ 0.02	0.5 $\pm$ 0.01	3.17 $\pm$ 0.01
ParaDiffUSE-EN	3.65 $\pm$ 0.29	11.34 $\pm$ 0.4	4.93 $\pm$ 0.26	2.17 $\pm$ 0.03	0.6 $\pm$ 0.01	3.27 $\pm$ 0.02

Table 3: Average performance on VB-DMD under matched conditions. Values are reported as mean $\pm$ standard error.

	SI-SDR $\pm$ SE	SI-SIR $\pm$ SE	SI-SAR $\pm$ SE	PESQ $\pm$ SE	ESTOI $\pm$ SE	DNS-MOS $\pm$ SE
Input	8.45 $\pm$ 0.2	8.45 $\pm$ 0.2	47.51 $\pm$ 0.36	3.02 $\pm$ 0.02	0.79 $\pm$ 0.01	3.07 $\pm$ 0.01
SGMSE+	17.16 $\pm$ 0.12	28.89 $\pm$ 0.19	17.65 $\pm$ 0.12	3.51 $\pm$ 0.02	0.85 $\pm$ 0.0	3.51 $\pm$ 0.01
Conv-TasNet	19.26 $\pm$ 0.12	32.36 $\pm$ 0.27	19.8 $\pm$ 0.12	3.4 $\pm$ 0.02	0.85 $\pm$ 0.0	3.31 $\pm$ 0.01
RemixIT	1.58 $\pm$ 0.09	9.22 $\pm$ 0.29	3.32 $\pm$ 0.05	2.13 $\pm$ 0.01	0.64 $\pm$ 0.0	2.9 $\pm$ 0.01
RVAE	17.03 $\pm$ 0.14	27.63 $\pm$ 0.24	17.85 $\pm$ 0.14	3.18 $\pm$ 0.02	0.81 $\pm$ 0.0	3.29 $\pm$ 0.01
UDiffSE	13.62 $\pm$ 0.14	20.08 $\pm$ 0.23	15.43 $\pm$ 0.12	3.34 $\pm$ 0.02	0.82 $\pm$ 0.0	3.31 $\pm$ 0.01
UDiffSE+	10.4 $\pm$ 0.11	22.45 $\pm$ 0.25	11.04 $\pm$ 0.08	3.3 $\pm$ 0.01	0.77 $\pm$ 0.0	3.27 $\pm$ 0.01
DEPSE-IL	10.48 $\pm$ 0.12	19.7 $\pm$ 0.24	11.54 $\pm$ 0.08	3.35 $\pm$ 0.02	0.78 $\pm$ 0.0	3.26 $\pm$ 0.01
DEPSE-TL	13.84 $\pm$ 0.16	16.36 $\pm$ 0.19	18.36 $\pm$ 0.12	3.41 $\pm$ 0.02	0.82 $\pm$ 0.0	3.31 $\pm$ 0.01
DiffUSEEN	14.32 $\pm$ 0.15	17.4 $\pm$ 0.2	18.56 $\pm$ 0.13	3.43 $\pm$ 0.02	0.83 $\pm$ 0.0	3.33 $\pm$ 0.01
ParaDiffUSE-IN	13.55 $\pm$ 0.14	20.72 $\pm$ 0.26	15.38 $\pm$ 0.1	3.41 $\pm$ 0.01	0.8 $\pm$ 0.0	3.33 $\pm$ 0.01
ParaDiffUSE-EN	17.85 $\pm$ 0.11	29.65 $\pm$ 0.2	18.41 $\pm$ 0.1	3.4 $\pm$ 0.02	0.84 $\pm$ 0.0	3.42 $\pm$ 0.01

Table 4: Average performance on VB-DMD under mismatched conditions. Values are reported as mean $\pm$ standard error.

	SI-SDR $\pm$ SE	SI-SIR $\pm$ SE	SI-SAR $\pm$ SE	PESQ $\pm$ SE	ESTOI $\pm$ SE	DNS-MOS $\pm$ SE
Input	8.45 $\pm$ 0.2	8.45 $\pm$ 0.2	47.51 $\pm$ 0.36	3.02 $\pm$ 0.02	0.79 $\pm$ 0.01	3.07 $\pm$ 0.01
SGMSE+	10.06 $\pm$ 0.18	12.43 $\pm$ 0.22	17.13 $\pm$ 0.26	3.29 $\pm$ 0.02	0.82 $\pm$ 0.0	3.3 $\pm$ 0.01
Conv-TasNet	11.13 $\pm$ 0.14	15.22 $\pm$ 0.2	15.9 $\pm$ 0.2	3.28 $\pm$ 0.01	0.83 $\pm$ 0.0	3.27 $\pm$ 0.01
RemixIT	1.17 $\pm$ 0.09	7.04 $\pm$ 0.23	3.38 $\pm$ 0.05	2.1 $\pm$ 0.01	0.63 $\pm$ 0.0	2.86 $\pm$ 0.01
RVAE	12.77 $\pm$ 0.17	37.64 $\pm$ 0.38	12.85 $\pm$ 0.17	3.11 $\pm$ 0.01	0.76 $\pm$ 0.0	3.28 $\pm$ 0.01
UDiffSE	14.08 $\pm$ 0.11	21.66 $\pm$ 0.2	15.35 $\pm$ 0.09	3.31 $\pm$ 0.02	0.81 $\pm$ 0.0	3.29 $\pm$ 0.01
UDiffSE+	10.87 $\pm$ 0.06	28.27 $\pm$ 0.2	11.02 $\pm$ 0.06	3.3 $\pm$ 0.01	0.77 $\pm$ 0.0	3.26 $\pm$ 0.01
DEPSE-IL	11.14 $\pm$ 0.06	24.32 $\pm$ 0.2	11.53 $\pm$ 0.06	3.33 $\pm$ 0.02	0.78 $\pm$ 0.0	3.23 $\pm$ 0.01
DEPSE-TL	13.85 $\pm$ 0.15	17.15 $\pm$ 0.21	17.43 $\pm$ 0.1	3.34 $\pm$ 0.02	0.81 $\pm$ 0.0	3.28 $\pm$ 0.01
DiffUSEEN	14.61 $\pm$ 0.14	19.06 $\pm$ 0.21	17.28 $\pm$ 0.11	3.35 $\pm$ 0.02	0.82 $\pm$ 0.0	3.3 $\pm$ 0.01
ParaDiffUSE-IN	8.52 $\pm$ 0.18	10.04 $\pm$ 0.22	15.79 $\pm$ 0.12	3.26 $\pm$ 0.01	0.77 $\pm$ 0.0	3.26 $\pm$ 0.01
ParaDiffUSE-EN	10.71 $\pm$ 0.19	12.82 $\pm$ 0.24	17.65 $\pm$ 0.19	3.34 $\pm$ 0.02	0.83 $\pm$ 0.0	3.35 $\pm$ 0.01

A.2 Average metrics by noise type

We report the average performance of each method separately for each noise type in the WSJ0-QUT and VB-DMD test sets. This analysis helps assess whether the relative behavior of NMF-based and diffusion-based noise priors depends on the acoustic noise condition.

Table 5: Average SI-SDR per noise type in WSJ0-QUT matched conditions.

Metrics Noise types	SI-SDR				Avg
	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	
Input	-1.8	-3.84	-2.63	-2.23	-2.6
SGMSE+	6.57	11.69	7.5	8.42	8.53
Conv-TasNet	10.91	15.43	12.1	12.18	12.63
RemixIT	3.01	5.75	3.82	4.03	4.14
RVAE	2.81	9.9	4.82	7.35	6.22
UDiffSE	4.12	6.02	3.27	3.93	4.35
UDiffSE+	3.23	4.86	2.07	2.7	3.24
DEPSE-IL	3.33	4.58	2.16	2.84	3.25
DEPSE-TL	2.99	5.37	2.46	3.23	3.53
DiffUSEEN	3.96	8.88	3.85	4.78	5.37
ParaDiffUSE-IN	2.83	-0.06	2.74	2.53	2.02
ParaDiffUSE-EN	7.04	11.32	7.73	7.94	8.49

Table 6: Average ESTOI per noise type in WSJ0-QUT matched conditions.

Metrics Noise types	ESTOI				Avg
	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	
Input	0.46	0.56	0.5	0.48	0.5
SGMSE+	0.71	0.84	0.75	0.73	0.76
Conv-TasNet	0.78	0.9	0.82	0.81	0.83
RemixIT	0.63	0.76	0.67	0.66	0.68
RVAE	0.56	0.74	0.63	0.62	0.64
UDiffSE	0.59	0.69	0.6	0.59	0.61
UDiffSE+	0.56	0.64	0.56	0.55	0.58
DEPSE-IL	0.57	0.65	0.58	0.57	0.59
DEPSE-TL	0.58	0.67	0.6	0.59	0.61
DiffUSEEN	0.63	0.75	0.64	0.64	0.67
ParaDiffUSE-IN	0.55	0.54	0.58	0.55	0.56
ParaDiffUSE-EN	0.69	0.83	0.72	0.7	0.73

Table 7: Average DNS-MOS per noise type in WSJ0-QUT matched conditions.

Metrics Noise types	DNS-MOS				Avg
	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	
Input	2.54	3.12	2.64	2.48	2.69
SGMSE+	3.53	3.89	3.61	3.51	3.63
Conv-TasNet	3.41	3.86	3.51	3.49	3.57
RemixIT	2.88	3.35	3.03	2.98	3.06
RVAE	3.12	3.54	3.17	3.18	3.25
UDiffSE	2.98	3.35	3.02	2.92	3.06
UDiffSE+	3.13	3.43	3.14	3.07	3.19
DEPSE-IL	3.02	3.42	3.07	2.96	3.11
DEPSE-TL	2.92	3.41	2.98	2.86	3.04
DiffUSEEN	3.07	3.46	3.11	3.01	3.16
ParaDiffUSE-IN	3.24	3.34	3.22	3.17	3.24
ParaDiffUSE-EN	3.28	3.84	3.37	3.29	3.44

Table 8: Average SI-SDR per noise type in WSJ0-QUT mismatched conditions.

Metrics Noise types	SI-SDR				Avg
	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	
Input	-1.8	-3.84	-2.63	-2.23	-2.6
SGMSE+	0.92	5.66	2.74	2.91	3.02
Conv-TasNet	4.08	7.99	5.05	6.2	5.83
RemixIT	2.04	4.87	3.3	3.28	3.35
RVAE	2.13	8.37	3.51	5.88	4.98
UDiffSE	2.3	1.7	1.29	1.3	1.66
UDiffSE+	1.06	-0.18	-0.4	-0.33	0.06
DEPSE-IL	1.29	-0.37	-0.14	0.08	0.24
DEPSE-TL	2.52	3.99	1.84	2.24	2.66
DiffUSEEN	3.48	7.24	3.32	4.3	4.59
ParaDiffUSE-IN	0.43	-0.27	-0.15	-1.51	-0.39
ParaDiffUSE-EN	2.8	5.86	3.77	2.33	3.65

Table 9: Average ESTOI per noise type in WSJ0-QUT mismatched conditions.

Metrics Noise types	STOI				Avg
	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	
Input	0.46	0.56	0.5	0.48	0.5
SGMSE+	0.5	0.73	0.62	0.61	0.61
Conv-TasNet	0.57	0.71	0.62	0.63	0.63
RemixIT	0.59	0.74	0.65	0.63	0.65
RVAE	0.53	0.68	0.59	0.57	0.59
UDiffSE	0.55	0.59	0.56	0.54	0.56
UDiffSE+	0.52	0.56	0.52	0.5	0.52
DEPSE-IL	0.53	0.57	0.53	0.51	0.53
DEPSE-TL	0.57	0.63	0.59	0.57	0.59
DiffUSEEN	0.61	0.71	0.62	0.62	0.64
ParaDiffUSE-IN	0.48	0.53	0.51	0.48	0.5
ParaDiffUSE-EN	0.53	0.69	0.61	0.58	0.6

Table 10: Average DNS-MOS per noise type in WSJ0-QUT mismatched conditions.

Metrics Noise types	DNS-MOS				
	CAFE-CAFE-2	CAR-WINUPB-2	HOME-LIVINGB-2	STREET-KG-2	Avg
Input	2.54	3.12	2.64	2.48	2.69
SGMSE+	3.48	3.75	3.61	3.55	3.59
Conv-TasNet	3.0	3.21	3.04	3.03	3.07
RemixIT	2.81	3.21	2.93	2.84	2.94
RVAE	3.0	3.35	3.05	3.03	3.1
UDiffSE	3.03	3.29	3.03	2.94	3.07
UDiffSE+	3.1	3.35	3.07	3.01	3.13
DEPSE-IL	2.99	3.34	3.0	2.9	3.05
DEPSE-TL	2.95	3.39	2.99	2.89	3.05
DiffUSEEN	3.16	3.51	3.16	3.1	3.23
ParaDiffUSE-IN	3.09	3.34	3.14	3.12	3.17
ParaDiffUSE-EN	3.06	3.57	3.25	3.21	3.27

Table 11: Average SI-SDR per noise type in VB-DMD matched conditions.

Metrics Noise types	SI-SDR					Avg
	bus	cafe	living	office	psquare	
Input	8.35	8.5	8.47	8.46	8.45	8.45
SGMSE+	17.86	15.37	15.86	19.25	17.49	17.16
Conv-TasNet	19.46	17.58	17.94	21.56	19.75	19.26
RemixIT	2.11	1.76	1.25	0.16	2.62	1.58
RVAE	18.0	14.22	16.46	20.02	16.45	17.03
UDiffSE	11.72	13.02	13.04	15.86	14.47	13.62
UDiffSE+	9.09	10.29	10.25	11.46	10.9	10.4
DEPSE-IL	9.0	10.63	10.51	11.16	11.12	10.48
DEPSE-TL	11.11	13.4	13.67	16.33	14.68	13.84
DiffUSEEN	11.26	13.68	14.11	17.51	15.01	14.32
ParaDiffUSE-IN	13.24	13.2	13.33	12.96	14.99	13.55
ParaDiffUSE-EN	18.14	16.35	16.79	19.94	18.04	17.85

Table 12: Average ESTOI per noise type in VB-DMD matched conditions.

Metrics Noise types	STOI					Avg
	bus	cafe	living	office	psquare	
Input	0.86	0.69	0.72	0.89	0.77	0.79
SGMSE+	0.89	0.79	0.82	0.9	0.84	0.85
Conv-TasNet	0.88	0.8	0.82	0.9	0.84	0.85
RemixIT	0.66	0.61	0.62	0.67	0.64	0.64
RVAE	0.86	0.72	0.79	0.88	0.79	0.81
UDiffSE	0.87	0.75	0.78	0.88	0.8	0.82
UDiffSE+	0.81	0.72	0.74	0.83	0.77	0.77
DEPSE-IL	0.83	0.73	0.75	0.84	0.78	0.78
DEPSE-TL	0.87	0.75	0.78	0.89	0.81	0.82
DiffUSEEN	0.87	0.77	0.79	0.89	0.82	0.83
ParaDiffUSE-IN	0.85	0.75	0.76	0.86	0.8	0.8
ParaDiffUSE-EN	0.88	0.79	0.81	0.89	0.83	0.84

Table 13: Average DNS-MOS per noise type in VB-DMD matched conditions.

Metrics Noise types	DNS-MOS					
	bus	cafe	living	office	psquare	Avg
Input	3.27	2.87	2.82	3.32	3.08	3.07
SGMSE+	3.5	3.49	3.52	3.51	3.51	3.51
Conv-TasNet	3.35	3.24	3.27	3.38	3.32	3.31
RemixIT	2.93	2.89	2.85	2.9	2.92	2.9
RVAE	3.38	3.11	3.27	3.45	3.25	3.29
UDiffSE	3.42	3.2	3.19	3.45	3.3	3.31
UDiffSE+	3.32	3.2	3.21	3.35	3.25	3.27
DEPSE-IL	3.33	3.17	3.18	3.35	3.25	3.26
DEPSE-TL	3.4	3.21	3.2	3.45	3.31	3.31
DiffUSEEN	3.39	3.27	3.25	3.43	3.33	3.33
ParaDiffUSE-IN	3.39	3.23	3.26	3.44	3.33	3.33
ParaDiffUSE-EN	3.45	3.36	3.36	3.5	3.42	3.42

Table 14: Average SI-SDR per noise type in VB-DMD mismatched conditions.

Metrics Noise types	SI-SDR					
	bus	cafe	living	office	psquare	Avg
Input	8.35	8.5	8.47	8.46	8.45	8.45
SGMSE+	10.89	10.08	9.45	8.75	11.13	10.06
Conv-TasNet	10.8	11.93	11.38	9.03	12.51	11.13
RemixIT	1.38	1.48	1.15	-0.14	1.95	1.17
RVAE	13.53	10.28	12.45	15.78	11.85	12.77
UDiffSE	14.86	12.99	12.97	15.15	14.42	14.08
UDiffSE+	11.33	10.11	10.24	11.7	10.97	10.87
DEPSE-IL	11.5	10.57	10.57	11.75	11.32	11.14
DEPSE-TL	14.55	13.25	13.27	13.49	14.71	13.85
DiffUSEEN	15.67	13.78	13.77	14.46	15.39	14.61
ParaDiffUSE-IN	8.13	9.07	8.69	7.69	9.0	8.52
ParaDiffUSE-EN	10.63	11.21	10.64	8.49	12.55	10.71

Table 15: Average ESTOI per noise type in VB-DMD mismatched conditions.

Metrics Noise types	STOI					
	bus	cafe	living	office	psquare	Avg
Input	0.86	0.69	0.72	0.89	0.77	0.79
SGMSE+	0.88	0.75	0.77	0.89	0.81	0.82
Conv-TasNet	0.87	0.78	0.79	0.88	0.82	0.83
RemixIT	0.65	0.6	0.6	0.66	0.63	0.63
RVAE	0.8	0.67	0.75	0.84	0.73	0.76
UDiffSE	0.87	0.74	0.77	0.88	0.8	0.81
UDiffSE+	0.81	0.71	0.73	0.82	0.76	0.77
DEPSE-IL	0.82	0.72	0.74	0.84	0.77	0.78
DEPSE-TL	0.86	0.74	0.77	0.88	0.8	0.81
DiffUSEEN	0.86	0.75	0.77	0.88	0.81	0.82
ParaDiffUSE-IN	0.81	0.72	0.75	0.82	0.77	0.77
ParaDiffUSE-EN	0.87	0.77	0.8	0.89	0.83	0.83

Table 16: Average DNS-MOS per noise type in VB-DMD mismatched conditions.

Metrics Noise types	DNS-MOS					Avg
	bus	cafe	living	office	psquare	
Input	3.27	2.87	2.82	3.32	3.08	3.07
SGMSE+	3.43	3.21	3.12	3.42	3.32	3.3
Conv-TasNet	3.37	3.2	3.14	3.34	3.3	3.27
RemixIT	2.91	2.85	2.79	2.87	2.85	2.86
RVAE	3.36	3.15	3.25	3.4	3.26	3.28
UDiffSE	3.4	3.17	3.12	3.46	3.28	3.29
UDiffSE+	3.32	3.18	3.19	3.36	3.24	3.26
DEPSE-IL	3.3	3.16	3.13	3.34	3.21	3.23
DEPSE-TL	3.38	3.18	3.14	3.43	3.28	3.28
DiffUSEEN	3.36	3.22	3.17	3.43	3.3	3.3
ParaDiffUSE-IN	3.33	3.15	3.21	3.35	3.27	3.26
ParaDiffUSE-EN	3.43	3.27	3.25	3.45	3.36	3.35

A.3 Average metrics by noise type and SNR

We further decompose the results by both noise type and signal-to-noise ratio (SNR). This provides a more detailed view of how each method behaves across different acoustic conditions and noise levels.

Table 17: Average performance per noise type and SNR in WSJ0-QUT matched conditions. Best results per column (per noise type) are in bold.

Noise type	Method	SI-SDR $\uparrow$				ESTOI $\uparrow$				DNS-MOS $\uparrow$			
		-5	0	5	Avg	-5	0	5	Avg	-5	0	5	Avg
CAFE-CAFE-2	Input	-6.68	-1.40	3.36	-1.80	0.31	0.46	0.62	0.46	2.33	2.52	2.79	2.54
	SGMSE+	1.82	7.32	11.09	6.57	0.57	0.74	0.85	0.71	3.37	3.54	3.71	3.53
	Conv-TasNet	6.99	11.43	14.78	10.91	0.66	0.80	0.89	0.78	3.15	3.44	3.67	3.41
	RemixIT	2.26	3.46	3.31	3.01	0.54	0.65	0.71	0.63	2.76	2.91	2.99	2.88
	RVAE	-2.71	3.63	8.16	2.81	0.39	0.58	0.73	0.56	3.00	3.11	3.26	3.12
	UDiffSE	-2.68	5.68	9.99	4.12	0.37	0.62	0.79	0.59	2.72	3.00	3.24	2.98
	UDiffSE+	-2.99	5.01	8.13	3.23	0.35	0.59	0.76	0.56	2.82	3.14	3.49	3.13
	DEPSE-IL	-3.24	5.08	8.67	3.33	0.35	0.60	0.77	0.57	2.71	3.02	3.37	3.02
	DEPSE-TL	-3.95	4.11	9.59	2.99	0.37	0.61	0.79	0.58	2.63	2.96	3.19	2.92
	DiffUSEEN	-2.79	5.14	10.24	3.96	0.42	0.68	0.83	0.63	2.82	3.08	3.33	3.07
	ParaDiffUSE-IN	-4.43	4.14	9.55	2.83	0.35	0.58	0.77	0.55	3.03	3.27	3.46	3.24
	ParaDiffUSE-EN	1.42	8.02	12.30	7.04	0.50	0.72	0.85	0.69	2.99	3.32	3.56	3.28
CAR-WINUPB-2	Input	-8.26	-3.41	1.52	-3.84	0.46	0.58	0.66	0.56	2.93	3.17	3.29	3.12
	SGMSE+	8.87	12.27	14.75	11.69	0.79	0.86	0.89	0.84	3.81	3.94	3.93	3.89
	Conv-TasNet	13.05	15.89	18.05	15.43	0.86	0.92	0.93	0.90	3.73	3.92	3.96	3.86
	RemixIT	7.22	5.50	4.09	5.75	0.75	0.78	0.75	0.76	3.33	3.38	3.33	3.35
	RVAE	7.02	10.24	13.32	9.90	0.64	0.77	0.83	0.74	3.34	3.60	3.75	3.54
	UDiffSE	1.33	7.03	11.04	6.02	0.55	0.73	0.82	0.69	3.12	3.39	3.61	3.35
	UDiffSE+	0.61	6.02	9.14	4.86	0.50	0.69	0.79	0.64	3.18	3.52	3.66	3.43
	DEPSE-IL	-0.41	5.98	9.56	4.58	0.50	0.70	0.80	0.65	3.15	3.52	3.67	3.42
	DEPSE-TL	-0.33	6.58	11.52	5.37	0.52	0.71	0.82	0.67	3.18	3.46	3.64	3.41
	DiffUSEEN	4.49	10.00	13.39	8.88	0.60	0.81	0.87	0.75	3.20	3.51	3.72	3.46
	ParaDiffUSE-IN	-6.75	1.05	7.51	-0.06	0.40	0.57	0.69	0.54	3.08	3.44	3.56	3.34
	ParaDiffUSE-EN	8.72	11.77	14.24	11.32	0.76	0.85	0.89	0.83	3.68	3.93	3.95	3.84
HOME-LIVINGB-2	Input	-7.18	-2.23	2.55	-2.63	0.38	0.49	0.64	0.50	2.49	2.63	2.82	2.64
	SGMSE+	3.22	8.24	11.98	7.50	0.65	0.76	0.86	0.75	3.51	3.62	3.71	3.61
	Conv-TasNet	8.80	12.57	15.66	12.10	0.73	0.83	0.90	0.82	3.35	3.53	3.70	3.51
	RemixIT	3.66	4.18	3.64	3.82	0.62	0.69	0.72	0.67	2.94	3.05	3.11	3.03
	RVAE	0.03	5.57	9.94	4.82	0.50	0.65	0.78	0.63	3.00	3.20	3.35	3.17
	UDiffSE	-2.83	4.49	9.52	3.27	0.44	0.62	0.78	0.60	2.81	3.05	3.25	3.02
	UDiffSE+	-3.94	3.71	7.77	2.07	0.40	0.59	0.75	0.56	2.87	3.20	3.40	3.14
	DEPSE-IL	-3.98	3.71	8.10	2.16	0.41	0.60	0.76	0.58	2.84	3.11	3.33	3.07
	DEPSE-TL	-3.84	3.51	9.14	2.46	0.44	0.62	0.78	0.60	2.77	3.00	3.22	2.98
	DiffUSEEN	-2.09	4.96	9.99	3.85	0.48	0.67	0.81	0.64	2.90	3.15	3.32	3.11
	ParaDiffUSE-IN	-3.28	3.62	9.23	2.74	0.43	0.58	0.76	0.58	2.98	3.24	3.50	3.22
	ParaDiffUSE-EN	3.22	8.23	12.77	7.73	0.58	0.75	0.86	0.72	3.16	3.40	3.59	3.37
STREET-KG-2	Input	-7.35	-2.45	2.76	-2.23	0.32	0.47	0.63	0.48	2.26	2.46	2.72	2.48
	SGMSE+	4.34	8.87	11.86	8.42	0.59	0.75	0.85	0.73	3.32	3.50	3.71	3.51
	Conv-TasNet	8.57	12.44	15.35	12.18	0.71	0.82	0.90	0.81	3.23	3.52	3.71	3.49
	RemixIT	4.32	4.35	3.51	4.03	0.59	0.67	0.72	0.66	2.86	3.02	3.06	2.98
	RVAE	2.30	8.05	11.51	7.35	0.43	0.64	0.78	0.62	2.91	3.16	3.45	3.18
	UDiffSE	-3.45	5.30	9.72	3.93	0.36	0.60	0.78	0.59	2.54	2.95	3.24	2.92
	UDiffSE+	-4.18	4.35	7.80	2.70	0.33	0.57	0.74	0.55	2.67	3.08	3.45	3.07
	DEPSE-IL	-4.27	4.35	8.26	2.84	0.35	0.58	0.76	0.57	2.52	2.99	3.34	2.96
	DEPSE-TL	-4.29	4.12	9.52	3.23	0.37	0.60	0.78	0.59	2.46	2.89	3.22	2.86
	DiffUSEEN	-2.51	6.05	10.58	4.78	0.42	0.67	0.82	0.64	2.64	3.06	3.33	3.01
	ParaDiffUSE-IN	-5.30	3.23	9.30	2.53	0.35	0.54	0.75	0.55	2.82	3.17	3.49	3.17
	ParaDiffUSE-EN	2.32	8.63	12.63	7.94	0.51	0.74	0.85	0.70	2.90	3.33	3.62	3.29

Table 18: Average performance per noise type and SNR in WSJ0-QUT mismatched conditions. Best results per column (per noise type) are in bold.

Noise type	Method	SI-SDR $\uparrow$				ESTOI $\uparrow$				DNS-MOS $\uparrow$			
		-5	0	5	Avg	-5	0	5	Avg	-5	0	5	Avg
CAFE-CAFE-2	Input	-6.68	-1.40	3.36	-1.80	0.31	0.46	0.62	0.46	2.33	2.52	2.79	2.54
	SGMSE+	-7.81	2.63	8.84	0.92	0.25	0.53	0.76	0.50	3.22	3.52	3.74	3.48
	Conv-TasNet	-2.04	5.02	9.95	4.08	0.35	0.59	0.79	0.57	2.67	3.01	3.36	3.00
	RemixIT	0.74	2.48	2.98	2.04	0.49	0.61	0.69	0.59	2.69	2.84	2.93	2.81
	RVAE	-3.38	2.96	7.45	2.13	0.36	0.54	0.71	0.53	2.87	3.00	3.17	3.00
	UDiffSE	-4.81	3.93	8.43	2.30	0.34	0.58	0.76	0.55	2.76	3.03	3.34	3.03
	UDiffSE+	-5.95	2.99	6.68	1.06	0.31	0.54	0.72	0.52	2.79	3.10	3.48	3.10
	DEPSE-IL	-5.68	2.98	7.17	1.29	0.32	0.55	0.73	0.53	2.67	2.98	3.36	2.99
	DEPSE-TL	-4.28	3.63	8.95	2.52	0.36	0.60	0.77	0.57	2.68	2.99	3.21	2.95
	DiffUSEEN	-3.08	4.67	9.55	3.48	0.40	0.65	0.81	0.61	2.92	3.17	3.42	3.16
	ParaDiffUSE-IN	-6.75	1.72	7.06	0.43	0.28	0.50	0.69	0.48	2.94	3.12	3.22	3.09
	ParaDiffUSE-EN	-4.94	4.48	9.61	2.80	0.30	0.57	0.75	0.53	2.81	3.10	3.29	3.06
CAR-WINUPB-2	Input	5.74	-3.41	1.52	-3.84	0.46	0.58	0.66	0.56	2.93	3.17	3.29	3.12
	SGMSE+	-1.33	8.02	12.15	5.66	0.58	0.79	0.84	0.73	3.57	3.83	3.90	3.75
	Conv-TasNet	3.60	9.14	12.48	7.99	0.56	0.76	0.84	0.71	2.81	3.30	3.65	3.21
	RemixIT	5.74	4.82	3.76	4.87	0.71	0.76	0.74	0.74	3.17	3.26	3.20	3.21
	RVAE	5.32	8.98	11.70	8.37	0.57	0.71	0.79	0.68	3.13	3.39	3.58	3.35
	UDiffSE	-3.16	1.69	8.14	1.70	0.46	0.59	0.76	0.59	3.05	3.29	3.62	3.29
	UDiffSE+	-5.45	0.15	6.42	-0.18	0.43	0.57	0.73	0.56	3.12	3.38	3.63	3.35
	DEPSE-IL	-6.11	0.02	6.78	-0.37	0.43	0.57	0.74	0.57	3.11	3.37	3.62	3.34
	DEPSE-TL	-1.80	3.85	11.84	3.99	0.48	0.64	0.83	0.63	3.15	3.40	3.69	3.39
	DiffUSEEN	2.84	8.01	12.17	7.24	0.56	0.76	0.86	0.71	3.21	3.61	3.79	3.51
	ParaDiffUSE-IN	-4.59	0.38	4.69	-0.27	0.42	0.54	0.65	0.53	3.15	3.39	3.51	3.34
	ParaDiffUSE-EN	1.30	6.02	11.72	5.86	0.56	0.72	0.84	0.69	3.31	3.63	3.82	3.57
HOME-LIVINGB-2	Input	-7.18	-2.23	2.55	-2.63	0.38	0.49	0.64	0.50	2.49	2.63	2.82	2.64
	SGMSE+	-4.32	4.36	9.75	2.74	0.42	0.66	0.82	0.62	3.41	3.67	3.78	3.61
	Conv-TasNet	-0.65	6.51	10.54	5.05	0.44	0.65	0.81	0.62	2.76	3.06	3.37	3.04
	RemixIT	2.94	3.66	3.36	3.30	0.58	0.67	0.71	0.65	2.84	2.93	3.03	2.93
	RVAE	-1.19	4.08	8.69	3.51	0.45	0.60	0.74	0.59	2.87	3.06	3.27	3.05
	UDiffSE	-5.10	2.25	8.16	1.29	0.41	0.57	0.75	0.56	2.82	3.03	3.29	3.03
	UDiffSE+	-7.05	1.00	6.33	-0.40	0.37	0.52	0.71	0.52	2.81	3.10	3.36	3.07
	DEPSE-IL	-6.64	0.98	6.68	-0.14	0.38	0.53	0.72	0.53	2.77	3.01	3.28	3.00
	DEPSE-TL	-4.55	2.50	9.02	1.84	0.43	0.59	0.77	0.59	2.77	3.03	3.24	2.99
	DiffUSEEN	-2.63	4.17	9.73	3.32	0.46	0.64	0.80	0.62	2.96	3.18	3.40	3.16
	ParaDiffUSE-IN	-7.00	0.70	7.38	-0.15	0.35	0.51	0.71	0.51	2.94	3.16	3.36	3.14
	ParaDiffUSE-EN	-2.83	4.76	10.86	3.77	0.43	0.63	0.80	0.61	2.99	3.32	3.50	3.25
STREET-KG-2	Input	-7.35	-2.45	2.76	-2.23	0.32	0.47	0.63	0.48	2.26	2.46	2.72	2.48
	SGMSE+	-5.20	4.20	9.45	2.91	0.38	0.65	0.80	0.61	3.33	3.57	3.74	3.55
	Conv-TasNet	0.98	6.94	10.50	6.20	0.43	0.64	0.80	0.63	2.71	2.98	3.38	3.03
	RemixIT	2.97	3.75	3.19	3.28	0.53	0.64	0.70	0.63	2.70	2.87	2.97	2.84
	RVAE	0.91	6.57	9.97	5.88	0.36	0.58	0.74	0.57	2.74	3.01	3.30	3.03
	UDiffSE	-6.77	2.22	8.11	1.30	0.33	0.53	0.74	0.54	2.56	2.95	3.29	2.94
	UDiffSE+	-8.24	0.89	6.06	-0.33	0.30	0.49	0.70	0.50	2.63	2.99	3.39	3.01
	DEPSE-IL	-7.60	1.01	6.49	0.08	0.31	0.50	0.71	0.51	2.48	2.89	3.29	2.90
	DEPSE-TL	-5.72	2.93	9.13	2.24	0.36	0.57	0.77	0.57	2.49	2.90	3.25	2.89
	DiffUSEEN	-3.12	5.66	10.12	4.30	0.40	0.65	0.80	0.62	2.79	3.11	3.38	3.10
	ParaDiffUSE-IN	-10.14	-0.85	6.02	-1.51	0.29	0.46	0.67	0.48	2.85	3.10	3.39	3.12
	ParaDiffUSE-EN	-7.50	4.23	10.50	2.33	0.33	0.60	0.80	0.58	2.84	3.21	3.55	3.21

Table 19: Average performance per noise type and SNR in VB-DMD matched conditions. Best results per column (per noise type) are in bold.

Noise type	Method	SI-SDR $\uparrow$					ESTOI $\uparrow$					DNS-MOS $\uparrow$				
		2.5	7.5	12.5	17.5	Avg	2.5	7.5	12.5	17.5	Avg	2.5	7.5	12.5	17.5	Avg
BUS	Input	0.89	5.99	10.92	15.95	8.35	0.78	0.87	0.89	0.92	0.86	3.10	3.24	3.33	3.39	3.27
	SGMSE+	16.59	17.65	17.98	19.30	17.86	0.85	0.90	0.91	0.93	0.89	3.50	3.51	3.51	3.49	3.50
	Conv-TasNet	17.93	19.03	19.62	21.36	19.46	0.83	0.89	0.90	0.92	0.88	3.22	3.35	3.39	3.43	3.35
	RemixIT	0.65	1.28	2.72	3.85	2.11	0.63	0.67	0.66	0.67	0.66	2.86	2.91	2.95	2.99	2.93
	RVAE	16.09	18.48	17.76	19.77	18.00	0.80	0.87	0.87	0.89	0.86	3.31	3.38	3.39	3.44	3.38
	UDiffSE	4.55	11.46	14.33	16.86	11.72	0.80	0.87	0.89	0.91	0.87	3.32	3.42	3.46	3.47	3.42
	UDiffSE+	2.94	9.52	11.50	12.66	9.09	0.75	0.82	0.83	0.84	0.81	3.22	3.32	3.36	3.38	3.32
	DEPSE-IL	2.68	9.15	11.54	12.87	9.00	0.76	0.83	0.85	0.86	0.83	3.21	3.34	3.39	3.40	3.33
	DEPSE-TL	4.24	9.94	13.56	17.02	11.11	0.81	0.88	0.89	0.91	0.87	3.31	3.42	3.42	3.47	3.40
	DiffUSEEN	4.72	10.18	13.55	16.88	11.26	0.81	0.88	0.89	0.91	0.87	3.29	3.40	3.40	3.46	3.39
	ParaDiffUSE-IN	8.28	13.83	14.77	16.28	13.24	0.81	0.86	0.86	0.87	0.85	3.32	3.39	3.41	3.45	3.39
	ParaDiffUSE-EN	14.96	18.85	19.16	19.72	18.14	0.83	0.88	0.89	0.91	0.88	3.38	3.47	3.46	3.50	3.45
CAFE	Input	-0.43	1.25	2.59	3.65	1.76	0.51	0.66	0.75	0.83	0.69	2.49	2.77	2.99	3.24	2.87
	SGMSE+	10.54	14.64	16.76	19.63	15.37	0.67	0.79	0.82	0.88	0.79	3.38	3.52	3.51	3.55	3.49
	Conv-TasNet	12.93	16.61	19.12	21.74	17.58	0.70	0.79	0.83	0.88	0.80	2.95	3.23	3.33	3.46	3.24
	RemixIT	-0.43	1.25	2.59	3.65	1.76	0.55	0.60	0.63	0.67	0.61	2.74	2.84	2.93	3.04	2.89
	RVAE	8.90	13.30	16.10	18.68	14.22	0.58	0.71	0.78	0.84	0.72	2.90	3.06	3.17	3.33	3.11
	UDiffSE	8.86	12.13	14.54	16.63	13.02	0.61	0.73	0.79	0.85	0.75	2.91	3.13	3.28	3.47	3.20
	UDiffSE+	7.58	10.05	11.32	12.26	10.29	0.59	0.70	0.76	0.81	0.72	2.95	3.17	3.27	3.41	3.20
	DEPSE-IL	7.61	10.48	11.81	12.69	10.63	0.60	0.72	0.77	0.82	0.73	2.92	3.15	3.24	3.39	3.17
	DEPSE-TL	7.97	12.08	15.36	18.31	13.40	0.62	0.74	0.80	0.86	0.75	2.90	3.16	3.30	3.48	3.21
	DiffUSEEN	8.61	12.35	15.42	18.40	13.68	0.65	0.75	0.81	0.86	0.77	3.02	3.23	3.33	3.48	3.27
	ParaDiffUSE-IN	8.41	12.60	15.18	16.71	13.20	0.61	0.74	0.79	0.84	0.75	2.99	3.20	3.31	3.42	3.23
	ParaDiffUSE-EN	12.06	15.72	17.91	19.80	16.35	0.68	0.78	0.81	0.86	0.79	3.18	3.37	3.41	3.50	3.36
LIVING	Input	1.07	6.12	10.97	16.07	8.47	0.57	0.72	0.76	0.85	0.72	2.51	2.73	2.93	3.15	2.82
	SGMSE+	11.87	15.70	16.83	19.21	15.86	0.73	0.83	0.84	0.89	0.82	3.46	3.54	3.53	3.55	3.52
	Conv-TasNet	14.00	17.56	18.96	21.40	17.94	0.74	0.82	0.83	0.88	0.82	3.05	3.24	3.35	3.43	3.27
	RemixIT	-1.89	0.62	2.49	3.90	1.25	0.57	0.63	0.63	0.65	0.62	2.72	2.79	2.92	2.97	2.85
	RVAE	12.17	16.05	17.74	20.04	16.46	0.70	0.79	0.81	0.87	0.79	3.07	3.25	3.35	3.41	3.27
	UDiffSE	8.81	12.61	14.46	16.43	13.04	0.67	0.78	0.80	0.86	0.78	2.89	3.14	3.33	3.39	3.19
	UDiffSE+	7.55	10.14	11.25	12.16	10.25	0.64	0.74	0.76	0.81	0.74	3.00	3.16	3.32	3.38	3.21
	DEPSE-IL	7.31	10.60	11.68	12.52	10.51	0.64	0.76	0.77	0.83	0.75	2.92	3.16	3.28	3.37	3.18
	DEPSE-TL	8.64	12.66	15.40	18.20	13.67	0.68	0.79	0.81	0.87	0.78	2.92	3.17	3.33	3.39	3.20
	DiffUSEEN	9.79	13.19	15.46	18.21	14.11	0.70	0.80	0.81	0.87	0.79	3.04	3.23	3.35	3.39	3.25
	ParaDiffUSE-IN	8.65	13.73	14.97	16.09	13.33	0.65	0.77	0.79	0.85	0.76	3.05	3.24	3.35	3.41	3.26
	ParaDiffUSE-EN	12.79	16.58	18.17	19.76	16.79	0.73	0.82	0.83	0.87	0.81	3.21	3.35	3.43	3.45	3.36
OFFICE	Input	0.87	5.90	10.93	15.91	8.46	0.83	0.86	0.91	0.94	0.89	3.11	3.30	3.40	3.47	3.32
	SGMSE+	17.37	18.70	19.70	21.15	19.25	0.87	0.89	0.91	0.94	0.90	3.52	3.51	3.53	3.49	3.51
	Conv-TasNet	19.69	20.86	22.06	23.57	21.56	0.86	0.88	0.91	0.93	0.90	3.20	3.37	3.47	3.49	3.38
	RemixIT	-3.59	-0.86	1.78	3.16	0.16	0.65	0.66	0.67	0.69	0.67	2.70	2.87	2.98	3.05	2.90
	RVAE	18.47	19.44	20.81	21.31	20.02	0.84	0.86	0.89	0.91	0.88	3.36	3.46	3.48	3.49	3.45
	UDiffSE	10.90	15.39	18.03	18.97	15.86	0.85	0.86	0.90	0.93	0.88	3.32	3.45	3.50	3.52	3.45
	UDiffSE+	8.31	11.73	12.76	12.93	11.46	0.79	0.81	0.84	0.86	0.83	3.28	3.37	3.36	3.37	3.35
	DEPSE-IL	6.77	11.33	13.09	13.28	11.16	0.81	0.83	0.86	0.88	0.84	3.28	3.36	3.38	3.40	3.35
	DEPSE-TL	11.61	15.32	18.27	19.97	16.33	0.85	0.87	0.90	0.92	0.89	3.35	3.46	3.48	3.49	3.45
	DiffUSEEN	14.45	16.38	18.90	20.20	17.51	0.86	0.87	0.90	0.92	0.89	3.33	3.45	3.46	3.48	3.43
	ParaDiffUSE-IN	6.25	11.96	16.39	16.98	12.96	0.83	0.84	0.87	0.89	0.86	3.37	3.45	3.47	3.45	3.44
	ParaDiffUSE-EN	18.77	19.59	20.25	21.11	19.94	0.86	0.88	0.91	0.93	0.89	3.39	3.49	3.55	3.55	3.50
PSQUARE	Input	0.90	6.04	10.86	15.90	8.45	0.66	0.75	0.82	0.85	0.77	2.78	3.03	3.21	3.28	3.08
	SGMSE+	14.91	16.57	18.58	19.88	17.49	0.78	0.83	0.87	0.88	0.84	3.42	3.55	3.55	3.51	3.51
	Conv-TasNet	16.94	18.65	20.95	22.42	19.75	0.77	0.83	0.87	0.88	0.84	3.09	3.34	3.45	3.40	3.32
	RemixIT	0.24	2.47	3.45	4.30	2.62	0.59	0.64	0.65	0.67	0.64	2.83	2.92	2.98	2.95	2.92
	RVAE	13.91	15.46	17.58	18.81	16.45	0.72	0.78	0.83	0.84	0.79	3.12	3.24	3.34	3.31	3.25
	UDiffSE	10.71	13.88	16.02	17.23	14.47	0.73	0.79	0.84	0.86	0.80	3.06	3.28	3.44	3.41	3.30
	UDiffSE+	8.53	10.81	11.90	12.32	10.90	0.70	0.76	0.80	0.81	0.77	3.06	3.25	3.38	3.32	3.25
	DEPSE-IL	8.13	11.28	12.38	12.66	11.12	0.71	0.77	0.82	0.82	0.78	3.05	3.25	3.37	3.31	3.25
	DEPSE-TL	9.94	13.50	16.46	18.76	14.68	0.74	0.80	0.85	0.86	0.81	3.09	3.31	3.45	3.39	3.31
	DiffUSEEN	10.74	13.88	16.62	18.75	15.01	0.76	0.81	0.85	0.86	0.82	3.14	3.33	3.44	3.39	3.33
	ParaDiffUSE-IN	11.95	15.08	16.35	16.53	14.99	0.74	0.80	0.83	0.84	0.80	3.18	3.34	3.42	3.38	3.33
	ParaDiffUSE-EN	15.56	17.37	19.27	19.93	18.04	0.77	0.83	0.86	0.87	0.83	3.28	3.45	3.49	3.45	3.42

Table 20: Average performance per noise type and SNR in VB-DMD mismatched conditions. Best results per column (per noise type) are in bold.

Noise type	Method	SI-SDR $\uparrow$					ESTOI $\uparrow$					DNS-MOS $\uparrow$				
		2.5	7.5	12.5	17.5	Avg	2.5	7.5	12.5	17.5	Avg	2.5	7.5	12.5	17.5	Avg
BUS	Input	0.89	5.99	10.92	15.95	8.35	0.78	0.87	0.89	0.92	0.86	3.10	3.24	3.33	3.39	3.27
	SGMSE+	5.81	8.20	12.38	17.47	10.89	0.81	0.88	0.90	0.92	0.88	3.35	3.43	3.45	3.50	3.43
	Conv-TasNet	7.21	8.83	11.87	15.51	10.80	0.81	0.87	0.89	0.92	0.87	3.28	3.37	3.40	3.45	3.37
	RemixIT	-0.54	0.31	2.19	3.65	1.38	0.61	0.66	0.66	0.67	0.65	2.83	2.90	2.93	2.98	2.91
	RVAE	12.02	13.93	13.59	14.63	13.53	0.75	0.81	0.81	0.84	0.80	3.28	3.38	3.37	3.42	3.36
	UDiffSE	11.78	14.35	15.85	17.61	14.86	0.81	0.87	0.88	0.91	0.87	3.30	3.40	3.43	3.45	3.40
	UDiffSE+	10.68	11.40	11.32	11.94	11.33	0.77	0.82	0.82	0.83	0.81	3.22	3.31	3.36	3.39	3.32
	DEPSE-IL	10.22	11.73	11.66	12.44	11.50	0.77	0.83	0.84	0.86	0.82	3.19	3.30	3.34	3.37	3.30
	DEPSE-TL	9.74	13.52	16.29	18.88	14.55	0.80	0.87	0.88	0.90	0.86	3.27	3.38	3.39	3.46	3.38
	DiffUSEEN	11.89	14.96	16.80	18.88	15.67	0.81	0.87	0.88	0.90	0.86	3.27	3.37	3.38	3.44	3.36
	ParaDiffUSE-IN	1.34	5.90	10.59	15.01	8.13	0.77	0.82	0.82	0.84	0.81	3.25	3.35	3.35	3.39	3.33
	ParaDiffUSE-EN	4.87	8.20	12.49	17.28	10.63	0.82	0.88	0.89	0.91	0.87	3.34	3.43	3.45	3.51	3.43
CAFE	Input	1.05	6.11	10.91	16.05	8.50	0.51	0.66	0.75	0.83	0.69	2.49	2.77	2.99	3.24	2.87
	SGMSE+	3.64	7.94	12.17	16.69	10.08	0.61	0.74	0.79	0.86	0.75	2.94	3.18	3.29	3.44	3.21
	Conv-TasNet	7.89	10.92	12.90	16.07	11.93	0.66	0.77	0.81	0.87	0.78	2.94	3.19	3.27	3.42	3.20
	RemixIT	-0.91	0.93	2.41	3.55	1.48	0.53	0.58	0.63	0.66	0.60	2.67	2.80	2.93	3.02	2.85
	RVAE	6.27	9.61	12.06	13.28	10.28	0.51	0.65	0.72	0.79	0.67	2.96	3.13	3.18	3.34	3.15
	UDiffSE	8.99	12.13	14.48	16.44	12.99	0.61	0.72	0.79	0.85	0.74	2.87	3.12	3.26	3.46	3.17
	UDiffSE+	8.11	9.78	11.00	11.60	10.11	0.59	0.70	0.76	0.81	0.71	2.94	3.15	3.26	3.39	3.18
	DEPSE-IL	8.35	10.35	11.47	12.17	10.57	0.60	0.71	0.77	0.82	0.72	2.91	3.13	3.22	3.38	3.16
	DEPSE-TL	7.45	12.02	15.40	18.25	13.25	0.60	0.73	0.79	0.86	0.74	2.85	3.14	3.29	3.46	3.18
	DiffUSEEN	8.24	12.78	15.70	18.50	13.78	0.63	0.74	0.80	0.86	0.75	2.93	3.19	3.29	3.47	3.22
	ParaDiffUSE-IN	2.26	7.42	11.50	15.24	9.07	0.58	0.72	0.77	0.82	0.72	2.84	3.16	3.25	3.37	3.15
	ParaDiffUSE-EN	4.84	9.55	13.47	17.10	11.21	0.65	0.77	0.81	0.87	0.77	3.02	3.27	3.32	3.48	3.27
LIVING	Input	1.07	6.12	10.97	16.07	8.47	0.57	0.72	0.76	0.85	0.72	2.51	2.73	2.93	3.15	2.82
	SGMSE+	2.35	7.29	11.94	16.55	9.45	0.64	0.77	0.80	0.86	0.77	2.86	3.04	3.22	3.36	3.12
	Conv-TasNet	7.05	10.17	12.80	15.69	11.38	0.68	0.80	0.82	0.87	0.79	2.87	3.09	3.27	3.34	3.14
	RemixIT	-1.95	0.51	2.35	3.81	1.15	0.54	0.62	0.62	0.64	0.60	2.67	2.74	2.85	2.92	2.79
	RVAE	9.24	13.92	12.81	13.89	12.45	0.64	0.76	0.76	0.82	0.75	3.07	3.22	3.33	3.39	3.25
	UDiffSE	8.87	12.52	14.38	16.25	12.97	0.65	0.77	0.79	0.86	0.77	2.80	3.06	3.29	3.35	3.12
	UDiffSE+	8.08	10.34	10.86	11.74	10.24	0.63	0.74	0.75	0.81	0.73	2.98	3.14	3.29	3.34	3.19
	DEPSE-IL	8.12	10.69	11.34	12.18	10.57	0.63	0.75	0.76	0.82	0.74	2.89	3.06	3.26	3.32	3.13
	DEPSE-TL	7.41	12.25	15.43	18.21	13.27	0.65	0.77	0.80	0.86	0.77	2.85	3.09	3.28	3.34	3.14
	DiffUSEEN	8.37	13.00	15.61	18.32	13.77	0.66	0.78	0.80	0.86	0.77	2.92	3.15	3.29	3.33	3.17
	ParaDiffUSE-IN	1.42	6.92	11.43	15.31	8.69	0.63	0.76	0.77	0.83	0.75	2.95	3.20	3.33	3.39	3.21
	ParaDiffUSE-EN	3.62	8.77	13.31	17.19	10.64	0.71	0.81	0.82	0.87	0.80	2.95	3.24	3.37	3.42	3.25
OFFICE	Input	0.87	5.90	10.93	15.91	8.46	0.83	0.86	0.91	0.94	0.89	3.11	3.30	3.40	3.47	3.32
	SGMSE+	1.66	6.19	11.06	15.85	8.75	0.84	0.87	0.90	0.93	0.89	3.26	3.41	3.49	3.52	3.42
	Conv-TasNet	2.97	6.83	11.11	14.99	9.03	0.83	0.86	0.91	0.94	0.88	3.13	3.31	3.43	3.48	3.34
	RemixIT	-3.91	-1.27	1.48	3.02	-0.14	0.63	0.65	0.66	0.69	0.66	2.63	2.84	2.96	3.04	2.87
	RVAE	15.12	16.24	15.37	16.37	15.78	0.81	0.83	0.84	0.87	0.84	3.32	3.37	3.44	3.48	3.40
	UDiffSE	11.56	14.63	16.39	17.92	15.15	0.85	0.86	0.90	0.93	0.88	3.32	3.47	3.50	3.53	3.46
	UDiffSE+	10.65	11.68	12.12	12.31	11.70	0.80	0.81	0.83	0.85	0.82	3.31	3.36	3.38	3.38	3.36
	DEPSE-IL	10.13	11.92	12.30	12.60	11.75	0.81	0.82	0.85	0.87	0.84	3.26	3.34	3.36	3.38	3.34
	DEPSE-TL	7.30	12.09	15.80	18.55	13.49	0.85	0.86	0.90	0.92	0.88	3.32	3.44	3.48	3.49	3.43
	DiffUSEEN	8.79	13.42	16.56	18.88	14.46	0.85	0.86	0.89	0.92	0.88	3.33	3.44	3.45	3.48	3.43
	ParaDiffUSE-IN	0.60	5.52	10.17	14.22	7.69	0.80	0.80	0.83	0.86	0.82	3.27	3.37	3.38	3.38	3.35
	ParaDiffUSE-EN	1.32	6.07	10.89	15.42	8.49	0.85	0.87	0.90	0.93	0.89	3.30	3.45	3.51	3.53	3.45
PSQUARE	Input	0.90	6.04	10.86	15.90	8.45	0.66	0.75	0.82	0.85	0.77	2.78	3.03	3.21	3.28	3.08
	SGMSE+	5.32	9.42	12.90	16.78	11.13	0.73	0.80	0.84	0.86	0.81	3.15	3.31	3.41	3.41	3.32
	Conv-TasNet	9.34	11.63	13.37	15.65	12.51	0.75	0.82	0.85	0.86	0.82	3.12	3.31	3.40	3.37	3.30
	RemixIT	-0.79	1.62	2.90	4.05	1.95	0.58	0.63	0.64	0.66	0.63	2.72	2.84	2.93	2.93	2.85
	RVAE	9.58	11.74	13.03	13.02	11.85	0.64	0.73	0.78	0.78	0.73	3.14	3.26	3.33	3.29	3.26
	UDiffSE	11.38	13.74	15.71	16.79	14.42	0.72	0.79	0.84	0.85	0.80	3.04	3.26	3.41	3.40	3.28
	UDiffSE+	9.96	10.71	11.61	11.59	10.97	0.70	0.76	0.80	0.80	0.76	3.08	3.25	3.36	3.28	3.24
	DEPSE-IL	9.99	11.20	12.10	12.00	11.32	0.71	0.76	0.81	0.81	0.77	3.03	3.20	3.31	3.29	3.21
	DEPSE-TL	9.46	13.66	16.85	18.78	14.71	0.73	0.79	0.84	0.85	0.80	3.06	3.28	3.42	3.38	3.28
	DiffUSEEN	10.92	14.46	17.23	18.89	15.39	0.74	0.86	0.84	0.85	0.81	3.09	3.31	3.42	3.36	3.30
	ParaDiffUSE-IN	1.91	7.11	11.69	15.18	9.00	0.70	0.77	0.80	0.81	0.77	3.07	3.30	3.38	3.34	3.27
	ParaDiffUSE-EN	6.12	11.37	14.93	17.67	12.55	0.77	0.82	0.86	0.86	0.83	3.19	3.36	3.47	3.43	3.36

B Effect of speech and noise mismatch

In this section, we investigate how speech and noise mismatches individually affect the performance of **ParaDiffUSE-EN** on the WSJ0-QUT and VoiceBank-DEMAND (VB-DMD) test sets. To this end, we use the separated version of **ParaDiffUSE-EN**, in which the speech and noise diffusion priors are trained as two separate models. This choice allows us to independently control whether the speech prior and the noise prior are matched or mismatched with respect to the test set, and therefore to better identify which source of mismatch is more detrimental to performance. In contrast, in the joint version, speech and noise are modeled by the same conditional score network, making it more difficult to attribute performance degradation to either the speech prior or the noise prior alone.

For each test set, we evaluate **ParaDiffUSE-EN** under four configurations:

- **Full match:** the speech diffusion model is trained on the speech dataset matching the test speech, and the noise diffusion model is trained on the noise dataset matching the test noise.
- **Speech mismatch only:** the speech diffusion model is trained on a speech dataset that does not match the test speech, while the noise diffusion model is trained on the noise dataset matching the test noise.
- **Noise mismatch only:** the speech diffusion model is trained on the speech dataset matching the test speech, while the noise diffusion model is trained on a noise dataset that does not match the test noise.
- **Full mismatch:** both the speech diffusion model and the noise diffusion model are trained on datasets that do not match the corresponding speech and noise components of the test set.

The results are reported in Tables 21 and 22. On the VB-DMD test set (Table 22), replacing the DEMAND-trained noise diffusion model with the QUT-trained one, while keeping the VB-trained speech model, leads to a large degradation in SI-SDR compared to the full-match setting: from 17.32 dB to 9.66 dB, corresponding to a relative drop of about 44%. On the WSJ0-QUT test set (Table 21), the noise-mismatch-only setting also degrades SI-SDR, from 9.19 dB to 6.47 dB, corresponding to a relative drop of about 29.5% compared to the full-match setting.

These results suggest that noise mismatch has a strong impact on **ParaDiffUSE-EN**. The degradation is particularly pronounced when evaluating on VB-DMD with a QUT-trained noise prior. One possible explanation is that the DEMAND training noises are more diverse than the QUT training noises. As a result, a DEMAND-trained noise model may generalize better to WSJ0-QUT than a QUT-trained noise model does to VB-DMD. This interpretation is consistent with the stronger performance drop observed on VB-DMD under noise mismatch.

On the WSJ0-QUT test set, speech mismatch and noise mismatch lead to SI-SDR degradations of a similar order. However, the underlying error profiles are different. Noise mismatch primarily affects the ability to remove interference, as reflected by the lower SI-SIR compared to the speech-mismatch-only setting (13.72 dB vs. 25.39 dB). The same pattern is observed more clearly on VB-DMD, where the SI-SIR is 11.83 dB under noise mismatch only, compared to 34.49 dB under speech mismatch only.

For perceptual metrics, the trends are less systematic. On WSJ0-QUT, compared to the full-match setting, PESQ appears less affected by noise mismatch than by speech mismatch, whereas DNS-MOS P.808 appears less affected by speech mismatch than by noise mismatch. However, this pattern is not clearly observed on VB-DMD. Overall, these results indicate that SI-SDR and SI-SIR are more directly sensitive to the type of mismatch, while PESQ and DNS-MOS exhibit more dataset-dependent behavior.

Table 21: Average performance of separated **ParaDiffUSE-EN** on WSJ0-QUT test set in different configurations.

	SI-SDR	SI-SIR	SI-SAR	PESQ	STOI	DNS-MOS
Input	-2.60 $\pm$ 0.16	-2.60 $\pm$ 0.16	46.66 $\pm$ 0.43	1.83 $\pm$ 0.02	0.50 $\pm$ 0.01	2.69 $\pm$ 0.01
speech_WSJ0_noise_QUT_full_match	9.19 $\pm$ 0.16	24.63 $\pm$ 0.25	9.41 $\pm$ 0.16	2.68 $\pm$ 0.02	0.76 $\pm$ 0.01	3.50 $\pm$ 0.01
speech_WSJ0_noise_DMD_only_noise_mismatch	6.47 $\pm$ 0.25	13.72 $\pm$ 0.4	8.40 $\pm$ 0.19	2.47 $\pm$ 0.02	0.68 $\pm$ 0.01	3.25 $\pm$ 0.02
speech_VB_noise_QUT_only_speech_mismatch	6.72 $\pm$ 0.15	25.39 $\pm$ 0.22	6.82 $\pm$ 0.15	2.31 $\pm$ 0.02	0.66 $\pm$ 0.01	3.44 $\pm$ 0.01
speech_VB_noise_DMD_full_mismatch	3.62 $\pm$ 0.29	11.05 $\pm$ 0.44	5.42 $\pm$ 0.24	2.28 $\pm$ 0.02	0.60 $\pm$ 0.01	3.23 $\pm$ 0.02

Table 22: Average performance of separated **ParaDiffUSE-EN** on VB-DMD test set in different configurations.

	SI-SDR	SI-SIR	SI-SAR	PESQ	STOI	DNS-MOS
Input	8.45 $\pm$ 0.20	8.45 $\pm$ 0.20	47.51 $\pm$ 0.36	3.02 $\pm$ 0.02	0.79 $\pm$ 0.01	3.07 $\pm$ 0.01
speech_VB_noise_DMD_full_match	17.32 $\pm$ 0.11	28.48 $\pm$ 0.18	17.9 $\pm$ 0.11	3.50 $\pm$ 0.01	0.84 $\pm$ 0.00	3.43 $\pm$ 0.01
speech_VB_noise_QUT_only_noise_mismatch	9.66 $\pm$ 0.18	11.83 $\pm$ 0.22	16.0 $\pm$ 0.15	3.36 $\pm$ 0.01	0.82 $\pm$ 0.00	3.38 $\pm$ 0.01
speech_WSJ0_noise_DMD_only_speech_mismatch	16.51 $\pm$ 0.10	34.49 $\pm$ 0.29	16.67 $\pm$ 0.09	3.34 $\pm$ 0.02	0.82 $\pm$ 0.00	3.37 $\pm$ 0.01
speech_WSJ0_noise_QUT_full_mismatch	10.59 $\pm$ 0.19	12.98 $\pm$ 0.23	16.66 $\pm$ 0.17	3.35 $\pm$ 0.01	0.83 $\pm$ 0.00	3.35 $\pm$ 0.01

C Performance–speed trade-off of DiffUSEEN and DEPSE-TL

In this section, we analyze how the performance of DiffUSEEN and DEPSE-TL evolves with the number of reverse diffusion iterations under mismatched conditions on the WSJ0-QUT and VB-DMD test sets. The experiment is conducted on an NVIDIA Tesla T4 GPU with 15 GiB of VRAM.

For mismatched WSJ0-QUT, DiffUSEEN achieves SI-SDR, PESQ, and ESTOI values that are greater than or equal to those of DEPSE-TL for all tested numbers of reverse iterations, except at 10 iterations. For mismatched VB-DMD, the decrease in SI-SDR and PESQ of DiffUSEEN after 30 iterations leads to a slight underperformance compared to DEPSE-TL. As expected, the real-time factor increases with the number of reverse iterations. However, this additional computational cost does not necessarily translate into substantial gains in PESQ or ESTOI for either method.

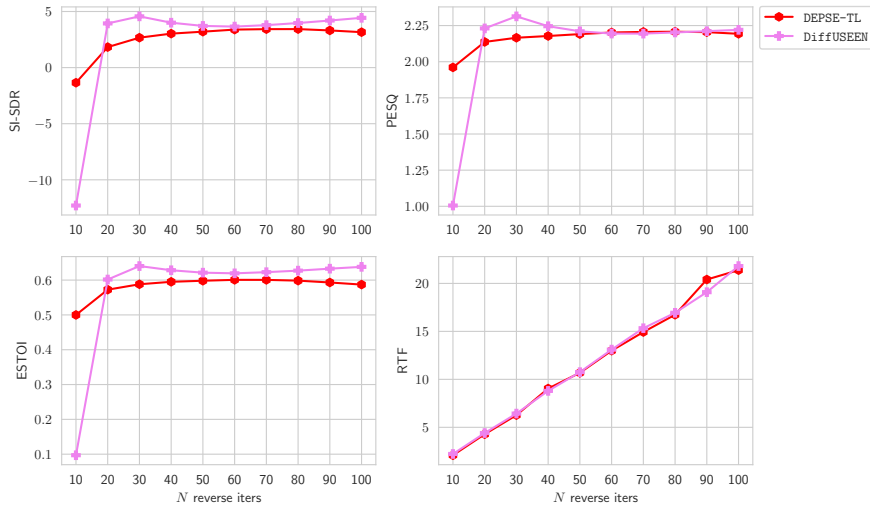


Figure 1: Performance and inference speed on the WSJ0-QUT test set under mismatched conditions as a function of the number of reverse diffusion steps N .

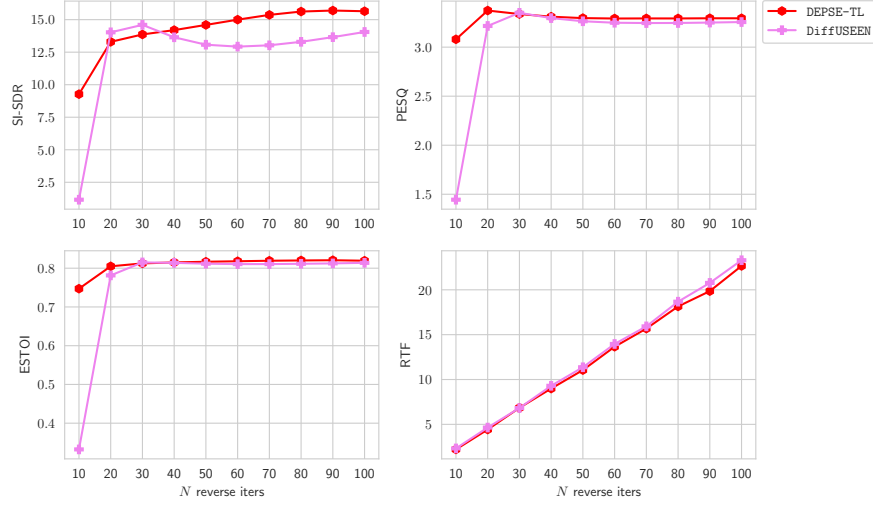


Figure 2: Performance and inference speed on the VB-DMD test set under mismatched conditions as a function of the number of reverse diffusion steps N .

D Metric distributions in matched and mismatched scenarios

We provide violin plots of the DNS-MOS, ESTOI, PESQ, and SI-SDR distributions for the different methods on the WSJ0-QUT and VB-DMD test sets, under both matched and mismatched conditions. For a given method, similar distributions across matched and mismatched scenarios indicate greater robustness to distribution shift and, therefore, better generalization to unseen test conditions, at least on the datasets considered here. A related plot, shown as Fig. 3 in the main paper, is analyzed in Section VI-B.2.

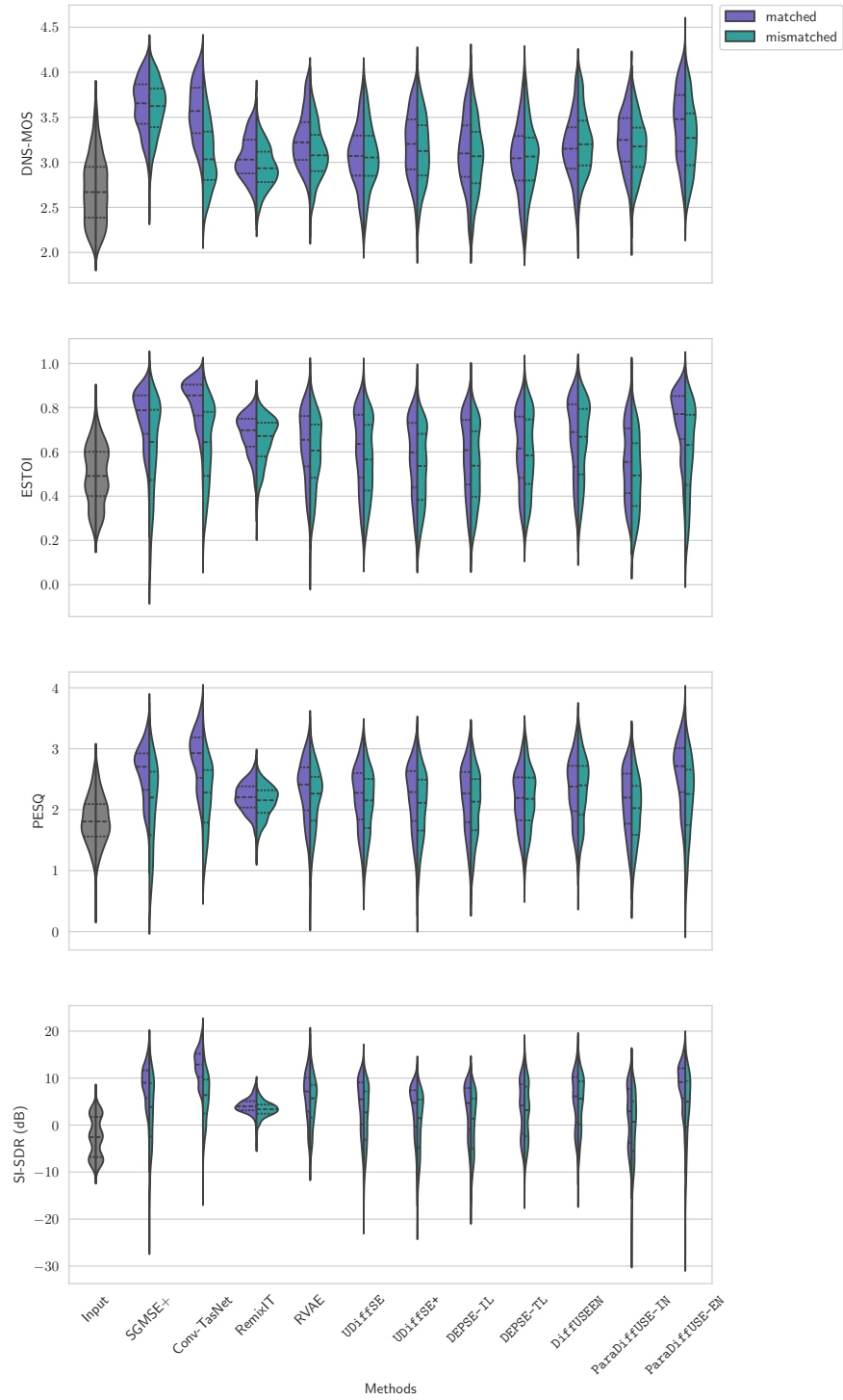


Figure 3: Violin plots showing the DNS-MOS, ESTOI, PESQ and SI-SDR distributions for the matched and mismatched conditions on the WSJ0-QUT test set, with dashed and dotted lines indicating the median and quartiles, respectively.

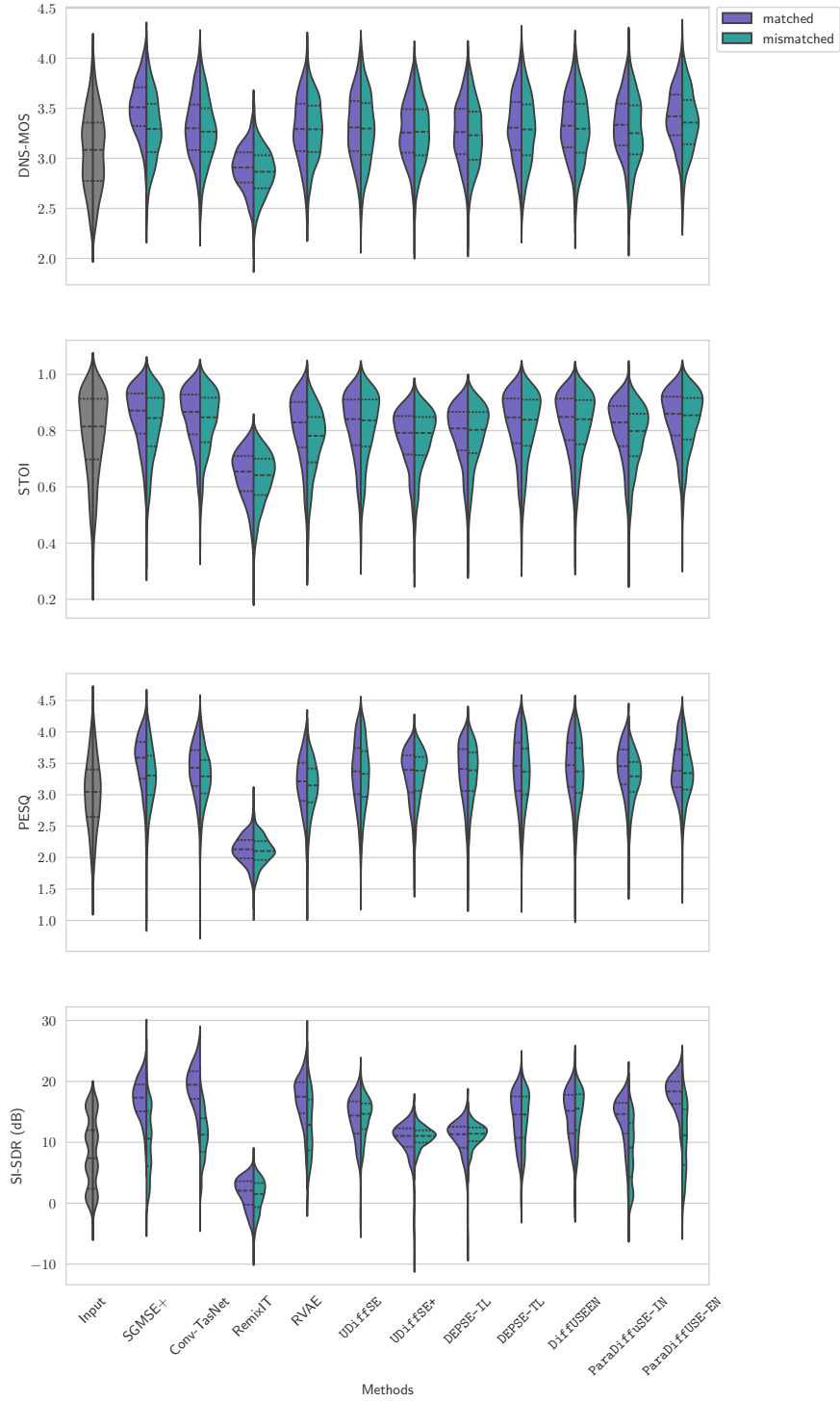


Figure 4: Violin plots showing the DNS-MOS, ESTOI, PESQ and SI-SDR distributions for the matched and mismatched conditions on the VB-DMD test set, with dashed and dotted lines indicating the median and quartiles, respectively.

E Preliminary trials on inference-time adaptation for the diffusion noise model in ParaDiffUSE-EN

In this section, we report preliminary experiments on adapting the diffusion noise model at inference time in the separated version of ParaDiffUSE-EN. The motivation is to mimic, in a diffusion-based noise-prior setting, the inference-time adaptation performed by NMF-based methods such as UDiffSE, DEPSE-TL, and DiffUSEEN, where the noise parameters are updated for each noisy utterance.

We consider the mismatched scenario in which evaluation is performed on the WSJ0-QUT test set, while the speech and noise priors are trained on the VB-DMD training set. In this setting, the diffusion noise model is initially trained on DEMAND noise and is therefore mismatched with respect to the QUT noise conditions encountered at test time.

Table 23: Experiment on fine-tuning the noise diffusion in inference time

	SI-SDR $\pm$ SE	SI-SIR $\pm$ SE	SI-SAR $\pm$ SE	PESQ $\pm$ SE	STOI $\pm$ SE
input	-2.55 $\pm$ 0.17	-2.55 $\pm$ 0.17	46.52 $\pm$ 0.44	1.84 $\pm$ 0.02	0.49 $\pm$ 0.01
ParaDiffUSE-EN	3.83 $\pm$ 0.29	11.67 $\pm$ 0.45	5.39 $\pm$ 0.25	2.28 $\pm$ 0.02	0.60 $\pm$ 0.01
ParaDiffUSE-EN_update_1	3.63 $\pm$ 0.29	11.65 $\pm$ 0.45	5.12 $\pm$ 0.25	2.26 $\pm$ 0.02	0.59 $\pm$ 0.01
ParaDiffUSE-EN_update_2	3.82 $\pm$ 0.29	11.73 $\pm$ 0.45	5.35 $\pm$ 0.25	2.28 $\pm$ 0.02	0.60 $\pm$ 0.01

After performing the posterior sampling of the noise, i.e. just after line 13 in ParaDiffUSE-EN algorithm in the main paper, we update the noise diffusion model parameters by using Adam optimizer in order to minimize the following losses:

- ParaDiffUSE-EN_update_1:

$$\mathcal{L}_1 = \|\sigma_{\tau_{i-1}} \mathbf{S}_{\theta}^n(\tilde{\mathbf{n}}_{0,i-1}, \tau_{i-1}) + \boldsymbol{\zeta}\|_2^2$$

where:

$$\tilde{\mathbf{n}}_{0,i-1} = \hat{\mathbf{n}}_{0,i-1} + \sigma_{\tau_{i-1}} \boldsymbol{\zeta}, \quad \boldsymbol{\zeta} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \mathbf{I})$$

- ParaDiffUSE-EN_update_2:

$$\mathcal{L}_2 = \|(\mathbf{x} - \hat{\mathbf{s}}_{0,i-1}) - \hat{\mathbf{n}}_{0,i-1}\|_2^2$$

In the above loss, $\hat{\mathbf{s}}_{0,i-1}$ and $\hat{\mathbf{n}}_{0,i-1}$ are computed using the Tweedie’s formula (see equation 11 in the main paper). Note that we got some out of memory issues when optimizing this losses on longer files, so we end up with the average performances computed on 590 files of WSJ0-QUT test set instead of performing the enhancement on 651 files and that is what is reported in the above table.

We can see that, with loss $\mathcal{L}_1$ the performances degrade a bit, while with $\mathcal{L}_2$ we just stay on par with the inference with no updating (ParaDiffUSE-EN).

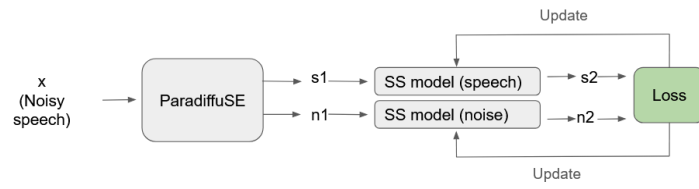
Other strategies investigated consisted (but with with less favourable results) in:

- using some lightweight Self-Supervised (SS) models (basically pretrained with masked spectrogram reconstruction task) for the speech and the noise respectively and fine-tune them during inference on the output of ParaDiffUSE. For finetuning the SS models we used a loss of the form following loss:

Loss = mixture consistency + dissimilarity enforcing

$$\mathcal{L}_3 = \|\mathbf{x} - (\mathbf{s}_2 + \mathbf{n}_2)\|^2 + \beta(-\|\mathbf{s}_2 - \mathbf{n}_2\|)^2$$

$$\mathcal{L}_4 = \|\mathbf{x} - (\mathbf{s}_2 + \mathbf{n}_2)\|^2 + \beta(\frac{1}{T} \sum_{t=1}^T \cos(\|\mathbf{s}_{2,t}\|, \|\mathbf{n}_{2,t}\|))$$



- Using a Deep Internal Learning framework instead of the SS models following the work : Ohnaka, Hien, and Ryoichi Miyazaki. "Unsupervised speech enhancement with spectral kurtosis and double deep priors." Acoustical Science and Technology 46.2 (2025): 146-156.